Fleet Rebalancing with Multi-Agent Reinforcement Learning under Passenger-Demand Shift: An NYC16 Study

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Abstract

We study whether a frozen multi-agent rebalancing policy keeps serving riders when demand moves away from its training distribution. The environment is NYC16: the 16 busiest taxi zones from real FHVHV demand, 26 drivers, 130 orders per hour, and five-minute rider patience. MAPPO is trained for 200 iterations on five seeds, frozen, and evaluated with deterministic argmax actions on eleven demand-shift scenarios (S0–S7) against a greedy dispatcher and a re-tuned rolling-horizon LP. At base demand MAPPO serves 64.2% ± 2.0 of orders at 397 ± 46 empty km per episode; the LP serves 69.7% ± 4.4 at 290 km, and greedy serves 44.8% ± 8.4 at 74 km. Every policy is most fragile when demand concentrates in one cold zone (S4): MAPPO retains 56.1% of its base service rate, the LP 67.3%, greedy 50.5%. The LP therefore leads the service/deadhead frontier at every shift, and MAPPO does not close that gap. IPPO is trained on the same protocol and reported alongside MAPPO. All numbers are five-seed means ± sample standard deviation.

1 Introduction

Problem. Fleet rebalancing is the decision of where to move idle drivers so that they meet demand before riders cancel. Rebalancing policies are trained on a demand distribution and deployed into one that moves. The study asks (RQ1) how much service rate degrades as demand shifts, and (RQ2) whether independent PPO and centralized-critic MAPPO degrade differently under the same shifts.

Environment. NYC16 is a 16-zone subset of Manhattan built from real FHVHV demand: 26 drivers, 130 orders per 60-minute window, four riders per driver, 2 km dispatch radius and five-minute order timeout. The demand source has 84 training, 12 validation and 12 test windows; validation and test windows are never used for training or checkpoint selection tuning.

Baselines. GreedyZoneDispatcher dispatches the nearest idle driver and never relocates. ZoneLPRebalancer is a rolling-horizon transportation LP with an EMA demand forecast, a five-minute lookahead and a deficit penalty, tuned on training windows only.

Method. MAPPO and IPPO share a zone actor with a 33-way discrete action space (idle, up to 16 dispatch slots, up to 16 relocation targets). The observation is the env-3.4 information set: 24 turns of demand history and an inbound-vehicle channel, 615 features per zone. Policies are trained for 200 iterations on five seeds and evaluated as frozen argmax policies. The team reward is

$$R = \gamma \cdot \text{served} - \beta \cdot \text{empty km} - \alpha \cdot \text{wait} - \text{cancel},$$

with $\alpha = 0.2$, $\beta = 0.06$, $\gamma = 1.0$ and a 0.65 relocator share carved out of γ, not paid on top of it.

Headline. MAPPO reaches 64.2% service at base demand, 5.5 points below the tuned LP and 27% above its deadhead. It retains 91.9% of base service on average across the ten shifted scenarios, but only 56.1% under cold-zone concentration. The LP is more robust on the same shift (67.3%). The supported claim is about the tradeoff frontier and its degradation, not that RL beats the baselines.

2 Environment and Baselines

2.1 NYC16

The 16 busiest taxi zones from real FHVHV data (zone IDs 68, 79, 112, 145, 161, 162, 163, 164, 170, 230, 231, 234, 236, 237, 246, 249), with centroids from the Manhattan road network. Each episode is a 60-minute demand window with $\mathrm{d}t = 0.5$ min, 26 drivers of capacity four, 130 orders, a 2 km dispatch radius and a five-minute timeout. The split is 84 train / 12 validation / 12 test windows.

2.2 Reward and observation

The served term pays 1.0 on completion. Empty kilometres are charged at $\beta = 0.06$ per km to the zone that commanded the move. Waiting is charged at $\alpha = 0.2$ per unit of waiting time to the zone containing the rider. A cancellation costs 1.0. The relocator share is 0.65 of the ride credit, split out of γ and shared along the relocation chain, so zone rewards sum to the team objective exactly (asserted by `tests/test_reward_sums_to_objective.py`).

2.3 Baselines

GreedyZoneDispatcher: nearest-idle dispatch, relocation disabled. **ZoneLPRebalancer**: rolling-horizon LP with EMA demand forecast (smoothing 0.3), five-minute lookahead and deficit penalty 1.0, re-tuned on NYC16 training windows. Both baselines are deterministic; their error bars in this report are the spread across the five evaluation windows, not across seeds.

3 Training

3.1 Protocol

MAPPO is trained for 200 iterations on seeds 0–4, six rollout episodes per iteration, four PPO epochs, learning rate 3×10^{-4}, $\gamma = 0.99$ and GAE $\lambda = 0.99$. Checkpoint selection uses the held-out validation split with 12 episodes at evaluation seed 9000; the selected checkpoint is the one reported. IPPO is trained on the identical protocol.

3.2 Final iteration

Arm	Service rate	Empty km / episode	Average wait
MAPPO-200	70.05% ± 2.55	457.6 ± 7.3	2.60 ± 0.12 min

These are on-policy training-rollout values, five-seed mean ± sample std. The plateau is reached before iteration 120: the five-seed mean service rate is 69.4% over iterations 101–120, 70.0% over 141–160 and 69.4% over 181–200. Training longer is not expected to move the final policy by more than about 1 point; the measured slope over iterations 101–200 is −0.16 points per 100 iterations.

3.3 Held-out validation

On 12 validation windows scored sequentially, the five-seed MAPPO arm serves 64.4% at 407.6 km and 2.73 min wait. The tuned LP scores 67.9% at 285.6 km on the identical windows. Paired per seed, MAPPO is -3.53 ± 1.10 points of service and $+122.0 \pm 18.0$ km of deadhead (mean ± standard error over five seeds); waiting time is a tie ($+0.00 \pm 0.04$ min). The information set does not close the gap to the LP.

4 Demand-Shift Evaluation

4.1 Scenarios

Eleven scenarios are evaluated, S0–S7. Magnitude shifts S2a–S2e scale the order cap by 0.5, 0.75, 1.5 and 2.0 relative to the 130-order base. S1 permutes demand spatially; S3 uses a validation-split window; S4 concentrates 60% of demand in cold zone 1; S5 puts 20% of demand into zone 0 between minutes 20 and 30; S6 uses the held-out test split; S7 drifts a hotspot from zone 12 to zone 10. S2c (magnitude 1.0) is not run because it is identical to S0.

4.2 Protocol

Every policy is frozen. Each scenario is evaluated on five demand windows, seeds 1000–1004, with deterministic argmax actions. An RL arm is evaluated on five training seeds over those five windows, so its mean is a five-seed mean and its error bar is the sample standard deviation across seeds. Greedy and LP are deterministic and are evaluated on the same five windows; their error bars are the spread across windows. The two error-bar definitions are not interchangeable.

4.3 Base demand

Policy	Service rate	Empty km / episode	Average wait
Greedy	44.8% ± 8.4	74	2.42 min
LP	**69.7%** ± 4.4	**290**	2.54 min
MAPPO-200	64.2% ± 2.0	397 ± 46	2.74 ± 0.14 min

MAPPO buys 19.4 points of service over greedy with 323 more empty kilometres, a factor of 5.4. The LP buys 5.5 more points than MAPPO while spending 107 fewer kilometres, so it dominates MAPPO on both axes at S0.

4.4 Degradation

Policy	S0	S4 cold zone	S6 held-out	Mean retention	Worst retention
Greedy	44.8%	22.6%	45.7%	91.5%	50.5%
LP	69.7%	46.9%	69.4%	**92.4%**	**67.3%**
MAPPO-200	64.2%	36.0%	65.7%	91.9%	56.1%

Retention is a policy's shifted service rate divided by its own S0 service rate; mean retention averages the ten shifted scenarios. MAPPO's degradation is concentrated in the two scenarios that change the spatial distribution rather than only its scale: S4 (36.0%, retention 56.1%) and S1, the spatial permutation (51.8%, retention 80.7%). Scalar volume shifts move it less: S2e (double demand) costs only 13.7 points, and S2b (0.75×) actually raises service slightly because fewer orders compete for the same fleet. Greedy and LP show the same qualitative pattern, and LP is the most robust at the hardest shift.

At S4, MAPPO serves 36.0% ± 1.4 and drives 478 ± 54 empty km; the LP serves 46.9% ± 4.0 at 280 km. The cold-zone shift is the case in which centralized rebalancing has the clearest advantage, and MAPPO does not capture it: its extra deadhead relative to the LP does not convert into service.

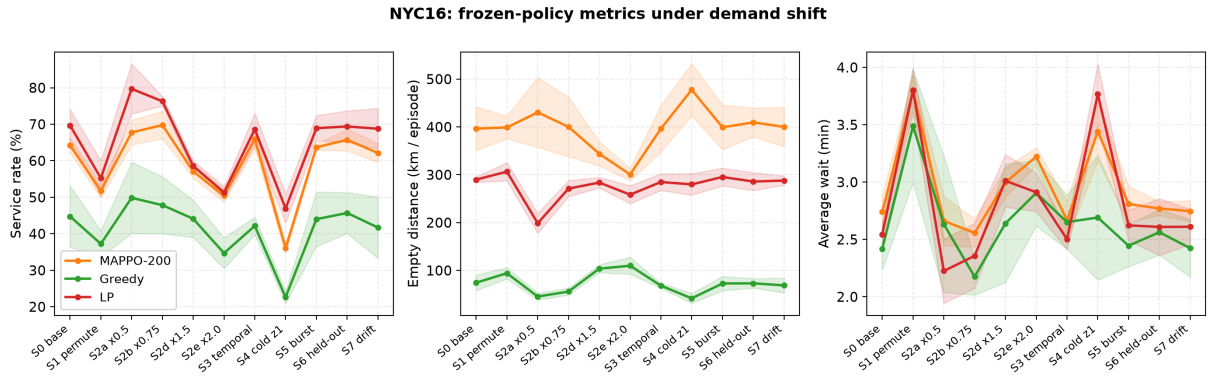


Figure 1: *Frozen-policy service rate, empty distance and average wait across S0–S7 on NYC16. Lines are means and bands are ±1 sample std: across five seeds for MAPPO, across five demand windows for the deterministic baselines.*

5 Discussion

5.1 The LP still leads the frontier

At base demand the LP is the only policy above 65% service, and it reaches that level with 27% less deadhead than MAPPO. At the hardest shift it keeps a 10.9-point service advantage at 198 fewer empty kilometres. The result is not that RL is useless: MAPPO beats greedy by 19.4 points at base demand and 13.4 points at S4. The result is that the learned relocation policy is less efficient per kilometre than a tuned centralized solver, and the gap widens where coordination matters most.

5.2 Where the degradation comes from

The scenarios that hurt MAPPO are spatial, not scalar. Doubling demand (S2e) costs 13.7 points, while permuting the same demand across zones (S1) costs 12.4 points and concentrating 60% of it

in one cold zone (S4) costs 28.2. A policy that has learned the training distribution's spatial layout loses more when that layout changes than when its volume changes. This is the deployment-relevant failure mode: the shift that matters is not a busier Friday, it is a new hotspot.

5.3 Why the gap does not close with more training

The five-seed training trace is flat from iteration 100 onward, and the destination logits span only a few tenths of a nat, so the policy has little ranking signal left to exploit. A supervised clone of the LP's actions into the same 33-way head reaches LP-level service in two minutes of CPU, which locates the gap in optimisation and credit assignment rather than in the actor's capacity. The clone is a diagnostic, not a method: it distils the baseline and must not be reported as a learned policy.

5.4 Limitations

- Five seeds is the minimum reportable unit; the shift-eval error bars describe seed-to-seed variation, not uncertainty over the full test split.
- Baselines are deterministic and evaluated on five windows, so their bars are window spread. RL-versus-baseline differences in Section 4 are not paired window-by-window; the paired comparison is the 12-validation-window result in Section 3.
- Policies are frozen after training and are never fine-tuned on a shifted scenario. S6 uses the held-out test split, but S1–S5 and S7 are constructed from the training split, so they measure robustness to transformations of seen windows, not to unseen time periods.
- One city, one fleet size, one demand source. No sim-to-real claim is made.

6 Reproducibility

All figures and tables are generated from artifacts in the repository:

- `runs/nyc16-h24-obs-s*/` – five-seed MAPPO checkpoints and per-iteration metrics
- `runs/nyc16-h24-obs-ippo-s*/` – five-seed IPPO checkpoints and per-iteration metrics
- `results/shift_eval_nyc/` – per-seed S0–S7 evaluations
- `results/shift_eval/summary.json` – NYC16 Greedy/LP rows
- `src/.../experiments/plot_shifts_v2.py` – all three figures

Figures are written to `results/figures/` and copied to `docs/figures/`. Code and artifacts: https://github.com/sharifmdathar/marl-fleet-rebalancing.

References

[1] X. Guo, N. S. Caros, and J. Zhao. Robust matching-integrated vehicle rebalancing in ride-hailing system with uncertain demand. *Transportation Research Part B: Methodological*, 150, 2021.

[2] J. K. Terry et al. PettingZoo: Gym for multi-agent reinforcement learning. *NeurIPS*, 2021. arXiv:2009.14471.